

IP-SAKTI Technical Documentation

AI Regulatory & Patent Intelligence Platform • Traditional Knowledge & Statutory Shayak

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Vercel

Executive Summary:

IP-SAKTI is an enterprise AI regulatory and patent intelligence platform designed to automate patent screening under the Indian Patent Act 1970, the Traditional Knowledge Digital Library (TKDL), the AYUSH Regulatory Corpus, and the Biological Diversity Act 2002. It combines high-recall statutory RAG grounding with real-time multi-stage reasoning to eliminate AI hallucinations in legal examination.

1. System Architecture & Core Capabilities

1.1 Problem Statement & Statutory Objectives

- Non-Patentability Exclusions (Section 3):** Automated evaluation against Section 3(p) (traditional knowledge exclusions), Section 3(d) (mere incremental modifications lacking enhanced therapeutic efficacy), and Section 3(e) (mere admixtures).
- Fact-Grounded Legal Answers:** Every synthesized output is accompanied by exact statutory citations, section extracts, confidence scores (98%+ alignment), and inspectable document snippets.
- Executive Reasoning Engine:** DeepSeek-R1 / ChatGPT-style multi-stage reasoning pipeline (Intent Interpretation → Statutory Archive Search → Section 3 Exclusion Screening → Synthesis) with live duration counters.
- Multimodal & Multilingual Access:** Supports document uploads (.pdf, .docx, .txt) with chunking efficiency visualization alongside regional Indian language voice assistant capabilities.

1.2 High-Level Architecture

Layer	Technologies & Components	Key Function
Frontend UI	Next.js 16 (App Router), React 19, TypeScript, Vanilla CSS	Responsive user dashboard, reasoning stream, citation modal & admin portal
AI / LLM Engine	Google Gemini 2.5 API (@google/genai), RAG Grounding Engine	Streaming natural language legal synthesis with strictly constrained grounding
Retrieval & RAG	Custom Token Dispersion Chunking Engine, Statutory Corpus	Pre-indexing of Indian Patent Act 1970, TKDL archives, and controller precedents
Database & ORM	MongoDB Atlas, Mongoose ORM	Stores user accounts, persistent chat histories, knowledge documents & audit logs
Security & Auth	NextAuth.js (JWT Strategy), bcrypt password hashing	Role-Based Access Control (User vs Admin), OTP password reset
App Distribution	Progressive Web App (PWA), Windows Desktop .zip Bundle	Offline service worker, standalone window mode, and desktop shortcut installer

2. Core Modules & Technical Specifications

2.1 User Intelligence Workspace (/dashboard)

- **Executive ChatGPT-Style Thinking Interface:** Displays the rotating IP-SAKTI App Logo spinner, live timer (e.g. 2.4s), stage badge (e.g. Stage 2/4), and linear progress indicator while reasoning. Collapses into a clean Thought for Xs accordion with zero emojis.
- **Verified Statutory Citations Grid:** Renders interactive citation cards containing document titles, section references, confidence match percentages, and highlighted legal excerpts with instant modal inspection.
- **Document Chunking & Token Dispersion Visualizer:** Allows users to attach patent drafts and inspect token efficiency, chunk boundaries, and lexical density.
- **Multilingual Voice Assistant:** Web Speech API powered microphone input and text-to-speech feedback.
- **RLHF Reinforcement System:** Thumbs-up/thumbs-down feedback recording with comment logging for model tuning.

2.2 Admin Telemetry & Control Portal (/admin/analytics)

- **System Metrics:** Live metrics for grounding accuracy (98.4%), average response latency (0.85s), token consumption, and query volume.
- **Knowledge Corpus Management:** Admin tools to upload, index, preview, and download statutory documents and guidelines.
- **User Management:** Role assignment, active session tracking, and user account provisioning.
- **Execution Audit Trail:** Immutable timestamped logs of all queries, document inspections, and downloads.

2.3 Conditional PWA & Windows App Distribution

- **PWA:** Includes offline Service Worker (/sw.js), Web Manifest (/manifest.json), and standalone app detection.
- **Windows Desktop App:** Standalone package with batch launcher (Launch-IP-SAKTI.bat), VBScript shortcut installer (Install-Desktop-Shortcut.vbs), and custom desktop icon (app.ico).
- **Conditional Hiding Logic:** Automatically removes the "Install" or "Desktop App" button once the user has installed or downloaded the app.

3. API Reference

Endpoint	Method	Auth	Description
/api/rag	POST	User / Admin	Core streaming RAG endpoint; returns real-time reasoning and grounded statutory response
/api/chats	GET, POST, DELETE	Session	Fetches, creates, or clears conversation threads and message histories
/api/documents	GET, POST, DELETE	Admin	Uploads, queries, downloads, or deletes statutory knowledge documents
/api/feedback	POST	User	Records user ratings (thumbs-up / thumbs-down) and comments in MongoDB

Endpoint	Method	Auth	Description
/api/stats	GET	Admin	Aggregates system-wide analytics, token latency, and query volume
/api/auth/signup	POST	Public	Registers new user accounts with bcrypt password hashing
/api/auth/forget-password	POST	Public	Dispatches 6-digit OTP to user email for password reset
/api/auth/reset-password	POST	Public	Verifies OTP and securely updates user password
/api/app/download	GET	Public	Streams the complete Windows Desktop Application zip installer package

4. Deployment & Repositories

- **Live Production URL:** <https://ip-sakti-seven.vercel.app>
- **GitHub Repository:** <https://github.com/danushkaviti-tech/IP-Shakthi-Shayak>
- **Documentation Download:** <https://ip-sakti-seven.vercel.app/downloads/IP-SAKTI-Documentation.pdf>